According to Lysenko, development of an annual seed plant consists of two phases. First phase is **thermostage**, which is vegetative phase requiring low temperature and suitable moisture. Next phase is **photo stage** which requires high temperature for synthesis of florigen (flowering hormone).

ii. Hypothesis of hormonal involvement

According to **Purvis** (1961), formation of a substance A from its precursor, is converted into B after chilling. The substance B is unstable. At suitable temperature B is converted into stable compound D called **Vernalin**. Vernalin is converted to F (Florigen). Florigen induces flower formation. At high temperature B is converted to C and devernalization occurs (Figure 15.11).

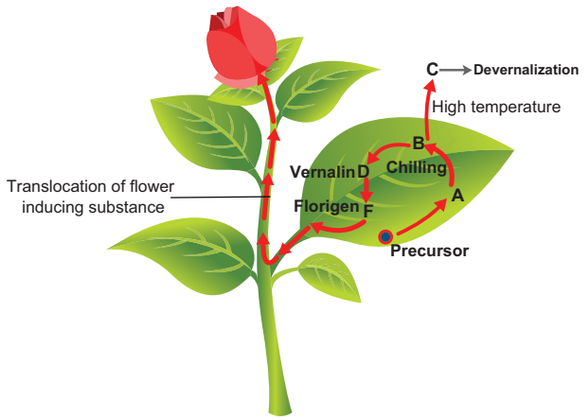


Figure 15.11: Vernalization and Flowering

2. Technique of Vernalization:

The seeds are first soaked in water and allowed to germinate at 10° C to 12° C. Then seeds are transferred to low temperature (3°C to 5°C) from few days to 30 days. Germinated seeds after this treatment are allowed to dry and then sown. The plants will show quick flowering when compared to untreated control plants.

3. Devernalization

Reversal of the effect of vernalization is called **devernalization**.

4. Practical applications

1. Vernalization shortens the vegetative period and induces the plant to flower earlier.
2. It increases the cold resistance of the plants.
3. It increases the resistance of plants to fungal disease.
4. Plant breeding can be accelerated.

15.5 Seed Germination and Dormancy

I. Seed Germination

The activation and growth of embryo from seed into seedling during favourable conditions is called **seed germination**.

1. Types of germination

There are two methods of seed germination. Epigeal and hypogeal.

i. Epigeal germination

During epigeal germination cotyledons are pushed out of the soil. This happens due to the elongation of the hypocotyl. Example: Castor and Bean.

ii. Hypogeal germination

During hypogeal germination cotyledons remain below the soil due to rapid elongation of epicotyls (Figure 15.12). Example: Maize, Pea.

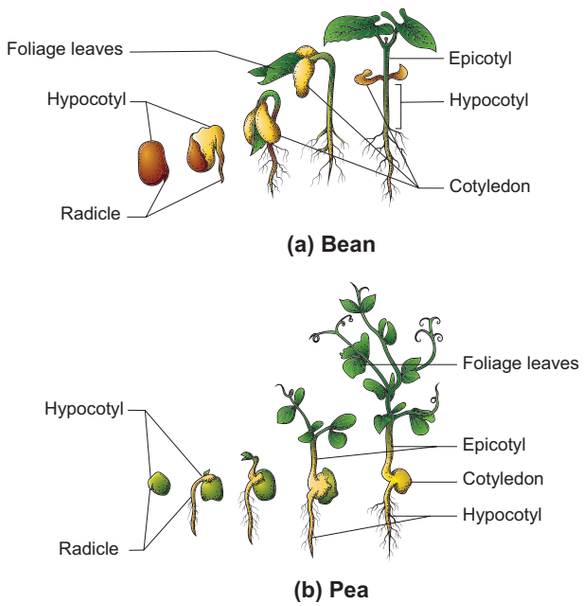


Figure 15.12: (a) Epigeal Germination
(b) Hypogeal Germination

2. Factors affecting germination

Seed germination is directly affected by external and internal factors:

i. External factors

a. **Water**: It activates the enzymes which digest the complex reserve foods of the

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